Project Proposal Template

Project:	Air Quality Forecasting and Public Health Early-Warning System
Objective:	To build an AI-driven machine learning pipeline and live public-facing early-warning platform that predicts 24–72 hour advance pollutant concentrations (PM2.5, PM10, NO2, O3, SO2, CO) and Air Quality Index (AQI) categories across Greater Cairo districts. The system leverages fused ground-station readings, satellite atmospheric data, and weather forecasts to enable proactive health advisories, automated subscriber alerts, and actionable mitigation for residents, schools, and health authorities before exposure occurs.
Description:	Air pollution is a severe, under-monitored public health threat causing respiratory and cardiovascular illness. This project addresses the gap by integrating multi-modal 100% open-access data sources—Open-Meteo Air Quality & Weather APIs, OpenAQ ground-station data, Copernicus Sentinel-5P (TROPOMI) satellite columns via Google Earth Engine, and OpenStreetMap geospatial layers. The end-to-end lifecycle spans automated data pipeline engineering, spatio-temporal ML forecasting (XGBoost/LightGBM/LSTM), MLOps experiment tracking (MLflow/DVC), and deployment via FastAPI, an interactive Streamlit map dashboard, and a Telegram alert bot.
Tools & Design: Technologies to Use:	• Programming & Core Tools: Python 3.10+, Jupyter Notebooks, Git & GitHub • Data Fusion & Processing: Pandas, NumPy, SciPy, OpenAQ API v3, Open-Meteo API, Google Earth Engine (EE Python API), Scikit-learn • Modeling & Analytics: XGBoost, LightGBM, Random Forest, Prophet, PyTorch / TensorFlow (LSTM/GRU), Optuna • MLOps & Experiment Tracking: MLflow, DVC (Data Version Control) • Deployment & Web Dashboard: FastAPI / Flask (REST API)/ Dash (Interactive Dashboard), Folium • Automated Alerts & Monitoring: Telegram Bot API / Webhooks, Evidently AI / Custom Monitoring Pipeline

Team Members:

1. Abdelrahman Rafiq Fathy	**4. Mohamed Amr Mohamed**
2. Abdullah Alaa Ali	**5. Yousif Mamdouh Hussein**
3. Abdelrahman Ashraf Mostafa	**6. Yousif Sherif Salah**

Milestone 1: Data Collection, Exploration, and Preprocessing (With Team Member Responsibilities)

- **Abdelrahman Rafiq Fathy — Open-Meteo Air Quality & Weather API Integration:** Build automated Python extraction scripts to pull historical and hourly multi-pollutant concentrations (PM2.5, PM10, NO2, O3, SO2, CO, UV index) and matching meteorological parameters (temperature, humidity, pressure, precipitation, wind speed/direction) for Greater Cairo districts.
- **Abdullah Alaa Ali — OpenAQ Ground Station Data Pipeline & Validation Setup:** Retrieve real ground-station sensor measurements via OpenAQ API v3 to establish baseline reference data; handle unit conversions (µg/m³ to ppb) and set up comparison routines between station data and model estimates.
- **Abdelrahman Ashraf Mostafa — Satellite & Geospatial Data Extraction (Copernicus Sentinel-5P & OSM):** Extract atmospheric column density layers (NO2, SO2, CO, O3) via Google Earth Engine (Sentinel-5P TROPOMI) and query road network/industrial zone boundaries using OSMnx for spatial feature building.
- **Mohamed Amr Mohamed — Data Cleaning, Time-Alignment & Outlier Filtering:** Implement missing value interpolation strategies (forward-fill/linear), align disparate data streams to a uniform 1-hour timestamp grid across coordinates, and apply statistical bounds to strip faulty sensor spikes.
- **Yousif Mamdouh Hussein — Exploratory Data Analysis (EDA) & Temporal Pattern Mining:** Analyze hourly, daily, weekly, and seasonal pollution cycles (rush hours, winter inversion layers, Khamsin dust season); produce time-series plots, correlation matrices, and wind-rose diagrams.
- **Yousif Sherif Salah — Feature Engineering (Time/Lag/Rolling) & EDA Report Compilation:** Engineer temporal features (hour, day of week, holiday/Ramadan flags) and rolling statistics (3h, 6h, 24h lag/moving averages); compile findings and visualization notebooks into the Milestone 1 EDA Report.

Deliverables:

✔ EDA Report: Comprehensive document detailing data sources, missing value strategies, pollution patterns, and weather interactions.

✔ Interactive EDA Notebooks: Jupyter notebooks featuring time-series plots, correlation heatmaps, and wind-rose diagrams.

✔ Cleaned & Time-Aligned Dataset: Merged, standardized hourly dataset (pollutants + weather + temporal/lag features) saved in Parquet/CSV format ready for modeling.

Milestone 2: Advanced Data Analysis and Spatial Feature Engineering

- Statistical & Causality Analysis: Conduct rigorous statistical hypothesis tests and correlation analyses to quantify weather drivers (wind speed/direction, temperature inversion layers, relative humidity) on PM2.5 and gas concentrations.
- Time-Series Decomposition: Perform additive/multiplicative decomposition on pollutant time series to isolate long-term trends, cyclic seasonality, and residual pollution spikes.
- Domain-Specific Feature Engineering: Construct domain features including rush-hour indicators, proximity to major highway networks and industrial corridors (from OSM), dust-season flags, and upwind/downwind directional vectors.
- Spatial Interpolation & Coverage Modeling: Implement spatial estimation algorithms (Inverse Distance Weighting / Kriging) combining sparse OpenAQ ground measurements with satellite grid data to estimate pollution at unmonitored district coordinates.
- Interactive Geospatial Heatmaps & Dashboards: Build interactive geospatial visualizations showing predicted pollution hotspots by neighborhood and district across Greater Cairo.

Deliverables:

✔ Data Analysis Report: Document outlining statistical driver findings, time-series decomposition results, and feature impacts.

✔ Enhanced Geospatial Heatmap Dashboard: Interactive map showcasing district-level pollution hotspots and contributing factors.

✔ Feature Engineering Summary & Matrix: Documentation of engineered spatial/domain features and expanded feature matrix for model training.

Milestone 3: Machine Learning Model Development, Tuning, and Evaluation

- Dual Problem Formulation: Frame modeling into two parallel targets: (1) Regression task predicting exact numerical concentrations for PM2.5, PM10, NO2, O3, SO2, CO; (2) Classification task categorizing EPA/WHO AQI health hazard levels (Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, Hazardous).
- Model Selection & Architecture Exploration: Implement and evaluate time-series baselines (SARIMA, Facebook Prophet), gradient boosted decision trees (XGBoost, LightGBM, Random Forest), and deep learning sequence architectures (LSTM/GRU).
- Time-Ordered Cross-Validation: Enforce strict rolling-window time-series validation splits to prevent future data leakage into historical training sets.
- Multi-Horizon Forecasting & Hyperparameter Optimization: Train and tune separate forecast models for 24-hour, 48-hour, and 72-hour prediction horizons using Bayesian Optimization / Optuna.
- Rigorous Model Evaluation & Cost-Sensitive Benchmarking: Measure regression performance (RMSE, MAE, MAPE) and classification performance (Precision, Recall, F1-Score), prioritizing high recall on 'Unhealthy' and 'Hazardous' categories to minimize un-alerted public health risks.

Deliverables:

✔ Model Evaluation & Comparison Report: Comparative performance matrix across regression and classification models, horizon benchmarks, and confusion matrices.

✔ Clean Model Code Base: Well-structured Python scripts/modules for data splitting, training, hyperparameter tuning, and evaluation.

✔ Serialized Final Models: Trained and tuned model artifacts (.pkl / .onnx) ready for pipeline deployment.

Milestone 4: MLOps Deployment, Live Monitoring, and Final Presentation

- MLOps Pipeline & Experiment Tracking: Integrate MLflow/DVC for model tracking, version control, and artifact storage to ensure full end-to-end reproducibility.
- RESTful API & Web Dashboard Deployment: Deploy model inference backend via FastAPI / Flask endpoints and build a live public Streamlit web dashboard displaying district forecasts, color-coded health maps, and actionable health guidance.
- Automated Public Health Alert System: Implement an automated Telegram Bot / Email Webhook notification system that broadcasts instant warnings to subscribers when district 24–72h forecasts cross into 'Unhealthy' thresholds.
- Real-Time Model Monitoring & Drift Detection: Establish automated monitoring comparing incoming live OpenAQ station data with model predictions to compute live accuracy metrics and trigger alerts upon performance degradation.
- Retraining Strategy & Final Deliverables: Define automated model retraining schedules for seasonal shifts (e.g., Khamsin dust season); complete the Final Comprehensive Project Report and deliver the live demo presentation.

Deliverables:

✔ Deployed Live Warning System: Production API endpoints, interactive Streamlit map dashboard, and active Telegram notification bot.

✔ MLOps & Monitoring Setup Documentation: Comprehensive guide detailing pipeline architecture, experiment tracking, and live drift detection setup.

✔ Final Project Report & Slide Deck: Complete technical project documentation and polished presentation deck for judges and stakeholders.